



Pixel Art on the Wall

A loop inside a loop draws a picture

반복문 안에 반복문 — 그림을 한 줄씩 그려요

BIG PICTURE

Pictures are made of rows



Inner loop = across (x). Outer loop = up (y).

안쪽 반복은 가로, 바깥쪽 반복은 위로

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a nested loop?

A loop **inside** another loop. The inner one finishes a whole row before the outer one steps up.

한 줄을 다 그린 다음 한 칸 위로 올라가요

CONCEPT

Inner vs outer



inner loop

grows x
(one row across)



outer loop

grows y
(next row up)

안쪽 = 가로 한 줄, 바깥쪽 = 다음 줄

CODE

A grid of red blocks

```
for y in range(2):           # 2 rows up
    for x in range(3):       # 3 across
        blocks.place(RED_WOOL, pos(x, y, 0))
```

3 blocks per row × 2 rows = **6 blocks** total.

CODE

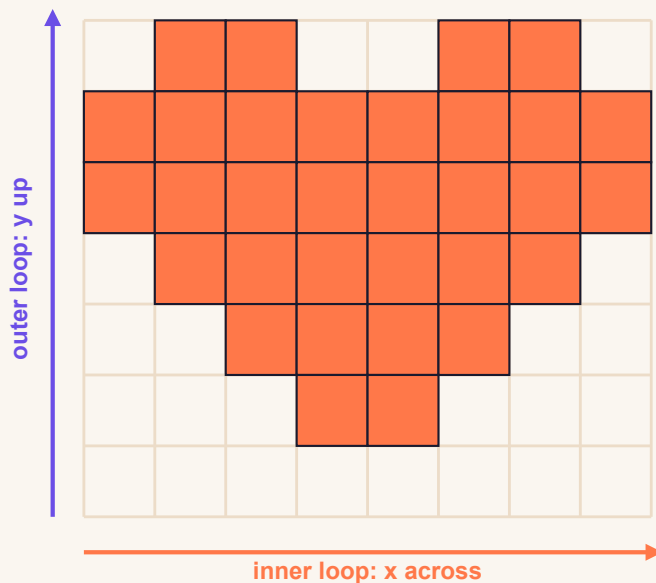
A patterned row: red, yellow, red

```
for y in range(3):  
    blocks.place(RED_WOOL, pos(0, y, 0))  
    blocks.place(YELLOW_WOOL, pos(1, y, 0))  
    blocks.place(RED_WOOL, pos(2, y, 0))
```

A yellow stripe runs straight up the middle ($x = 1$).

SEE IT

A picture, drawn row by row



안쪽 반복이 한 줄(x), 바깥쪽 반복이 다음 줄(y) — 하트가 완성돼요

PREDICT 🤔

How many blocks?

Predict first — then click reveal

```
y = 0
repeat 2 times:
  x = 0
  repeat 3 times:
    place red block at (x, y)
    add 1 to x
  add 1 to y
```

Reveal output 🔍 reveal

VOCAB

New words today

row a line of blocks across (one y level)

nested loop a loop inside another loop

inner the loop that runs across (x)

outer the loop that steps up (y)

줄 / 행

중첩 반복문

안쪽

바깥쪽



What's wrong with this code?

Goal: **3 rows** of blue, each one step higher.

buggy

```
y = 0
repeat 3 times:
  x = 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to x
```

Think first — then click next.



DEBUG

The bug

Bug: the outer loop never grows `y`, so all 3 rows stack on `y = 0`. Add `add 1 to y` at the end of the outer loop.

fixed

```
y = 0
repeat 3 times:
  x = 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to x
  add 1 to y
```

YOUR TURN!

Draw pixel art

- ✓ Plan a small picture: a heart, face, or your initial
- ✓ Use a nested loop — inner grows x, outer grows y
- ✓ Change block colours to make the pattern
- ✓ Make it at least 5 wide and 3 tall
- ✓ Send a photo of your art



QUIZ

QUESTION 1 OF 3

In a nested loop, what does the **inner** loop do?

Steps up to the next row (grows y).

Changes the block colour.

Nothing.

Draws one row across (grows x).



QUIZ

QUESTION 2 OF 3

Inner loop runs 4 times, outer loop runs 3 times. How many blocks?

4

12

3

7



QUIZ

QUESTION 3 OF 3

All your rows land in the same place. What did you forget?

Nothing — that's a feature.

To grow y in the outer loop.

To pick a colour.

To grow x in the inner loop.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Art on the wall!

Next: look down at the floor (x, z).

다음 주: 바닥을 내려다보기 (x, z)